

Irman Sazzad Selected Modeling Work

- **ERCOT Data Center Interconnection and Grid Resilience | Energy Systems, Infrastructure Finance, and Policy:** I designed an integrated framework for connecting large data center loads to ERCOT more quickly without shifting infrastructure costs, reliability risk, or price exposure onto existing ratepayers. I combined a capacity impact fee for transmission and stability upgrades, a long term geothermal power purchase agreement for price certainty, synchronous condensers for grid support, hydrogen fuel cells for clean backup, virtual power plant participation, and flexible demand obligations under Texas regulation. I evaluated interconnection, generation, resilience, financing, and regulatory compliance as one system, tying each technology and contract structure to a specific constraint while comparing alternatives through engineering requirements, operating costs, capital recovery, present value, and risk allocation. Related power system work required me to construct a least cost generation mix under capacity, firm availability, baseload, ramping, and peak demand constraints and assess how TVA Integrated Resource Plans responded to load growth, data centers, electrification, renewable integration, storage, reserve needs, and technology uncertainty. The analysis showed that the lowest cost technology was not necessarily the strongest system choice once fuel price correlation, outage exposure, curtailment, permitting, and interconnection timing were considered together. I ultimately framed accelerated interconnection as an investment and risk allocation problem in which developers internalize the grid costs they create while gaining price certainty and resilience. I also framed the PPA and broader proposal in nontechnical language for diverse stakeholders, balancing engineering feasibility, economic incentives, regulatory constraints, and political realities so the design remained practical and applicable.
- **U.S. Natural Gas Markets | Econometrics and Forecasting:** I examined Henry Hub price formation first as an economic relationship shaped by LNG exports, domestic production, storage, and international gas prices, then as a forecasting problem that had to perform on unseen data. Related energy markets work also applied financial derivatives and hedging strategies to commodity price risk. I assembled monthly EIA and international benchmark data in R and estimated log level, first difference, interaction, lagged, and predictive specifications using Newey West inference, Augmented Dickey Fuller tests, Engle Granger cointegration, Granger causality, and event based analysis. I then compared classical, X11, and STL decomposition, trend seasonal regression, ETS, seasonal ARIMA selected through exhaustive order search, and dynamic regression with crude oil prices inside an ARIMA error structure. I retained additional structure only when residual diagnostics and held out forecast performance justified it. The combined results supported partial integration between U.S. and international gas markets while favoring a parsimonious ARIMA structure for forecasting, reinforcing the distinction between explaining market relationships and predicting prices.
- **Oil Extraction Under Policy and Market Uncertainty | Optimal Control and Strategic Energy Market Analysis:** I examined oil extraction first as an intertemporal optimization problem and then as a strategic OPEC market decision under uncertainty. I formulated the producer's depletion problem from the reserve transition and profit objective, derived the Hamiltonian, costate equation, first order condition, transversality condition, and terminal stock rule using Pontryagin's Maximum Principle, and solved the system numerically in Mathematica to test how salvage incentives and discounting reshape extraction. I then moved the problem into an OPEC setting where finite reserves, production capacity, competing producers, and stochastic demand and supply shocks made prices incomplete signals of underlying behavior. I used market demand, residual supply, reserve scarcity, and evolving price information to reassess production, weighing immediate returns against preserving reserves for later periods. Together, the analyses showed how optimal depletion changes once strategic interaction, market power, imperfect information, and uncertainty enter the decision.
- **Renewable Energy Development | OLS, Machine Learning, Clustering, and Interactive Analysis:** I investigated which economic and energy system factors best explain cross country differences in renewable energy consumption. I built the data pipeline in Python and R by reconciling World Bank and Our World in Data panels, standardizing country definitions, units, and missing values, and conducting visual diagnostics before modeling. I estimated multivariate OLS and challenged the linear specification with random forest, support vector regression, held out evaluation, and k means clustering with standardized features and elbow based selection. I also built maps, interactive plots, and a Shiny interface for country and regional exploration. Agreement between linear and tree based models strengthened evidence around the main predictors, while disagreement identified relationships where nonlinear structure required closer scrutiny.
- **Nuclear Investment and Economic Development | Financial and Scenario Modeling:** I modeled how advanced nuclear and fuel cycle investments could translate into long run municipal property tax revenue and development capacity for Oak Ridge under defined planning assumptions. I built the Excel model at the company level using investment timing, declining taxable value schedules, and year specific tax rates before aggregating projects into a common horizon. I validated depreciation and tax assumptions and used scenario and sensitivity analysis to test how investment, incentive, and land use assumptions changed expected revenue and service capacity. The model gave city leadership a consistent basis for comparing development scenarios across the broader investment pipeline rather than one proposal at a time.

• **Pipeline Risk and Environmental Assurance | Principal Agent Contracting and Game Theory:** In my thesis, I modeled how a regulator can design environmental assurance when a pipeline operator's true safety cost and preventive effort are both unobservable. I formulated a principal agent problem with adverse selection and moral hazard, derived participation and incentive compatibility constraints, linked bond forfeiture to leak risk, and used R to search feasible contract menus and run comparative statics across damages, firm costs, revenue, landowner wealth, type shares, and risk preferences. The calibrated contracts differentiated requirements by firm type, induced preventive effort, and aligned with an existing federal bonding benchmark as an external policy check. I then built a companion strategic game in which the operator, regulator, and landowner chose actions under endogenous leak risk and asymmetric information; backward induction and programmed payoff tables produced an equilibrium where compliance, monitoring, and precaution were mutually optimal. Together, the models let me compare designed incentives with strategic behavior and assess how institutional design changes environmental risk.

• **Regional Energy Productivity Shock | Computable General Equilibrium Model:** I modeled how an energy productivity improvement propagates through an entire regional economy rather than evaluating the energy sector in isolation. I built and calibrated a multisector computable general equilibrium model in GAMS using a social accounting matrix, CES production and trade relationships, market clearing, and zero profit conditions, then solved the simultaneous system and stress tested progressively larger shocks across output, factor demand, prices, trade, household income, and welfare. The results showed higher energy output and household welfare alongside labor and capital reallocation across sectors, revealing tradeoffs that partial equilibrium analysis would miss. Related GAMS work extended the same approach into stochastic resource allocation using Monte Carlo simulation and uncertain profitability to test how risk changes optimal production decisions and the stability of simulated outcomes.

• **Data Centers and Housing Markets | Large Scale Geospatial Data Engineering:** I built the property level exposure data needed for a multistate hedonic analysis of how proximity to data centers may affect housing markets. I organized the R workflow into auditable stages for sales cleaning, assessor merges, facility cleaning, geocoding, coordinate repair, final coordinate selection, and nearest data center distance construction. DuckDB and Parquet handled very large files, while existing coordinates and ArcGIS, OpenStreetMap, and Census fallbacks were combined with source tracking and nondestructive outputs. I chunked distance calculations and embedded quality checks so exposure values could be traced to their source, while separating raw, intermediate, and final data and using Git based reproducibility practices so transformations could be rerun without altering original inputs. The result was an analysis ready dataset with transparent provenance and a defensible foundation for later causal and hedonic modeling.

• **Sales Tax Reporting Automation | Financial Data Workflow and Dashboard Automation:** I rebuilt a recurring sales tax reporting process whose manual structure created operational risk. I defined the business rules separating valid transactions from subtotals, grand totals, blanks, and other nontransaction rows, then implemented them consistently across PowerShell, Power Query, Excel, Power BI, and SharePoint. I designed the workflow to identify source county and reporting period, preserve original files, standardize outputs, and feed a repeatable dashboard infrastructure without changing the underlying financial logic. I required exact reconciliation with historical controls before treating the automation as complete, making preservation of financial meaning the acceptance criterion rather than automation alone. In my current role, I am extending this work into broader financial analysis and reporting, using SQL to prepare, automate, validate, and structure financial information for recurring reporting and decision support.

• **Technology Shocks and Macroeconomic Adjustment | RBC and DSGE Simulation:** I modeled how a stochastic technology shock propagates through household and firm decisions over time. I specified an RBC framework with optimizing households and firms, capital accumulation, labor supply, and an autoregressive productivity process, translated the nonlinear equilibrium conditions into Dynare, solved for the steady state, applied a local approximation, and generated impulse responses for consumption, investment, labor, output, capital, and productivity. I also distinguished broad productivity shocks from investment specific shocks that change the efficiency of new capital formation. The simulations showed how the same disturbance moves differently through consumption and investment, with investment responding more sharply as households and firms adjust intertemporally to changing productivity.

• **Microfinance Randomized Trial Replication and Extension | OLS, First Differences, IV, 2SLS:** I replicated the randomized evaluation of microfinance expansion by Banerjee, Duflo, Glennerster, and Kinnan and extended the analysis across later survey waves to test persistence and within household change. In Stata, I rebuilt baseline balance and outcome tables, applied survey weights and neighborhood clustered standard errors, estimated intent to treat effects with baseline controls, modeled first differences, and used randomized assignment as an instrument for actual take up in a two stage least squares extension. I treated successful replication as a prerequisite for extension, validating the empirical setup before introducing new specifications. The results suggested more persistent gains in productive assets than in profits or borrowing, while imprecise instrumental variable estimates were treated cautiously rather than oversold. Across my PhD level econometrics work, I also implemented fixed and random effects, GMM, logit and probit maximum likelihood, nonparametric and semiparametric regression, quantile regression, difference in differences, event studies, unit root and cointegration testing, and bootstrap inference in Stata and MATLAB, choosing estimators around the identification problem.